Making a Metamorphic Rock

You are now going to make a metamorphic chocolate rock. You will place different stresses on chocolate pieces to form a metamorphic chocolate rock. The chocolate pieces represent grains in a rock or more easily visualized, individual grains like sandstone. As different stresses are placed on pieces students can witness or imagine how real rock reacts to the same stresses.

Instructional Procedures:

1. Place glass container on heat source with heat source off.
2. Mix different kinds of chocolate pieces and M&M's and pour into glass container about 1 inch thick at bottom. Different colours allow students to choose a colour and watch as it changes instead of losing their 'grain' during the metamorphic process.
3. Explain that the different pieces represent minerals in a rock or different grains in a sedimentary rock, i.e. quartz, salt, feldspar, amphibole, sand.
4. Turn heat source on a low setting (do not let the mixture boil or melt). Warm the pieces until they become slightly soft.
5. Explain to kids that metamorphic rocks are rocks that have been changed from one rock type (sedimentary, metamorphic, igneous) into another rock type. The rock properties change like the chocolate pieces are changing before of them as they heat up. The minerals in real rock soften and are easily squished into different shapes when under pressure.
6. Place plastic wrap around the object to be pushed into the jar. Now place pressure onto the chocolate pieces.
7. This demonstration will take a few minutes to perform. That is fine, because time is the one factor affecting all metamorphic rock.
8. Explain to students that the combined effects of heat and pressure over long periods of time are what change other rock into metamorphic rocks.
9. Keep chocolate pieces on the heat source and under pressure until the pieces slowly squish together.
10. Once the mass has become one "newly formed rock" remove from heat. After it cools cut up into pieces and give to kids to examine and eat.

Discussion:

As you perform the demonstration ask students to list the different stresses that form metamorphic rocks. On the same paper have students draw what is happening to the chocolate pieces. First, the pieces are normal. Second, placed on heat. Third, pressure applied, etc. Ask if one of the forces were removed would a metamorphic rock form? Try removing one of the forces (heat, pressure, or time). How much did the rock change from all stresses compared to only one? What changed from the first picture they drew to the last picture of the chocolate rock.

Variation:

Instead of using chocolate based pieces you may want to try jellybeans or some type of jelly concoctions. Replace the condensed milk with a Karo syrup and water mixture. The demonstration can be performed the same way.